# Speech and Audio Systems

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

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WORKBOOK · 1 OF 21

## Recall and choose your starting point

> Record your starting explanation for Speech and Audio Systems before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Visual Composition

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Prompt Evaluation

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Examples and Rubrics

Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 21

## Practise with fading help

> Complete the supported Speech and Audio Systems trace and record the first step you can explain without help.

Use W2: Trace the worked case for Speech pipelines and multimodal **evidence**. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

## Individual reasoning check

> Check the proposed Speech and Audio Systems explanation against the supplied facts before discussing it.

Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 21

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

**Independent case: Changed independent case**

Known audio says “nineteen minus six”. **Transcription** says “ninety minus six”; the calculator returns84. Locate the first error and propose a check.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 21

## Your learning target

> Connect the Speech and Audio Systems target to a trace and a justified conclusion.

> Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

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WORKBOOK · 6 OF 21

## Solve the given case

> Solve the supplied Speech and Audio Systems case and show the deciding evidence.

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

Where did the error enter?

Show your trace, calculation or **evidence**

**Write or draw your reasoning and evidence.**

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WORKBOOK · 7 OF 21

## Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A recording says fifteen, but the transcript says fifty. The summary repeats fifty. Where did the error enter, and what should be checked first?

My independent answer and reason

**Write or draw your reasoning and evidence.**

WORKBOOK · 8 OF 21

## Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

WORKBOOK · 9 OF 21

## Project checkpoint

> Justify what Speech and Audio Systems adds to Verified Study Buddy and keep the evidence.

Show transcripts and require confirmation before saving memory or using tools.

The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

## Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

## Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

**My independently reasoned answer**

Record support used: none, hint or teacher explanation.

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WORKBOOK · 12 OF 21

## W1 Explain the mechanism

Explain the mechanism. Apply the supplied Speech and Audio Systems facts and keep your reasoning.

### Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Speech and Audio Systems", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**My reasoning and evidence**

WORKBOOK · 13 OF 21

## W2 Trace the worked case

> Trace the worked case. Apply the supplied Speech and Audio Systems facts and keep your reasoning.

### Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Speech and Audio Systems", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

WORKBOOK · 14 OF 21

## W3 Audit the limit

> Audit the limit. Apply the supplied Speech and Audio Systems facts and keep your reasoning.

A report omits this qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

WORKBOOK · 15 OF 21

## W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Speech and Audio Systems facts and keep your reasoning.

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

**My reasoning and evidence**

WORKBOOK · 16 OF 21

## W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Speech and Audio Systems facts and keep your reasoning.

### Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Speech and Audio Systems", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

WORKBOOK · 17 OF 21

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

## Pictorial case practice: Speech and Audio Systems

> A fictional learner dictates a calculation to a voice assistant. Explain the labelled step before checking the changed condition.

A fictional learner dictates a calculation to a voice assistant.

Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37.

Synthetic voice says “thirty-seven”. The example supplies ground truth; no real audio was analysed.

AI ACADEMY / AI-3/12

# Speech and Audio Systems

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Recognise speech

Transcription changes thirteen into thirty.



### 2. Calculate text

30+7=37 is correct for the mistaken transcript.



### 3. Speak result

Speech synthesis repeats 37; fluent voice does not validate the input.



### 4. Confirm intended input

Read back the recognised expression; intended 13+7=20.

**CHANGE ONE THING**

Correct the transcript to "thirteen plus seven" with all later stages unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |

| Evidence to keep | My record |
| --- | --- |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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WORKBOOK · 20 OF 21

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional learner dictates a calculation to a voice assistant.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Read back the recognised expression; intended 13+7=20.

> 1. Confirm intended **input**

2. Recognise speech

3. Calculate text

4. Speak result

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Confirm intended **input** Read back the recognised expression; intended 13+7=20. The first error occurs before arithmetic.

Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

### Practice 2: explain the deciding evidence

#### Changed case · predict first

Changed-case practice: Correct the transcript to “thirteen plus seven” with all later stages unchanged. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The calculator and spoken result should become 20 under the declared pipeline.

**My selected conclusion, deciding fact and explanation**

#### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: The calculator and spoken result should become 20 under the declared pipeline. 13+7=20; the changed **input** removes the earlier recognition error.

### Repair a missing building block

Review Visual Composition (ai-3/11). Goal: Explain the **mechanism** and terminology of image generation and provenance. Short practice: Which checks pass after that single change, and what still needs repair?.

Return to this week and explain what you changed.

Review Prompt Evaluation (ai-3/9). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Why is B's selection defensible here without calling it universally superior?.

Return to this week and explain what you changed.

Review Examples and Rubrics (ai-3/6). Goal: Write a prompt specification with observable response criteria. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card..

Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

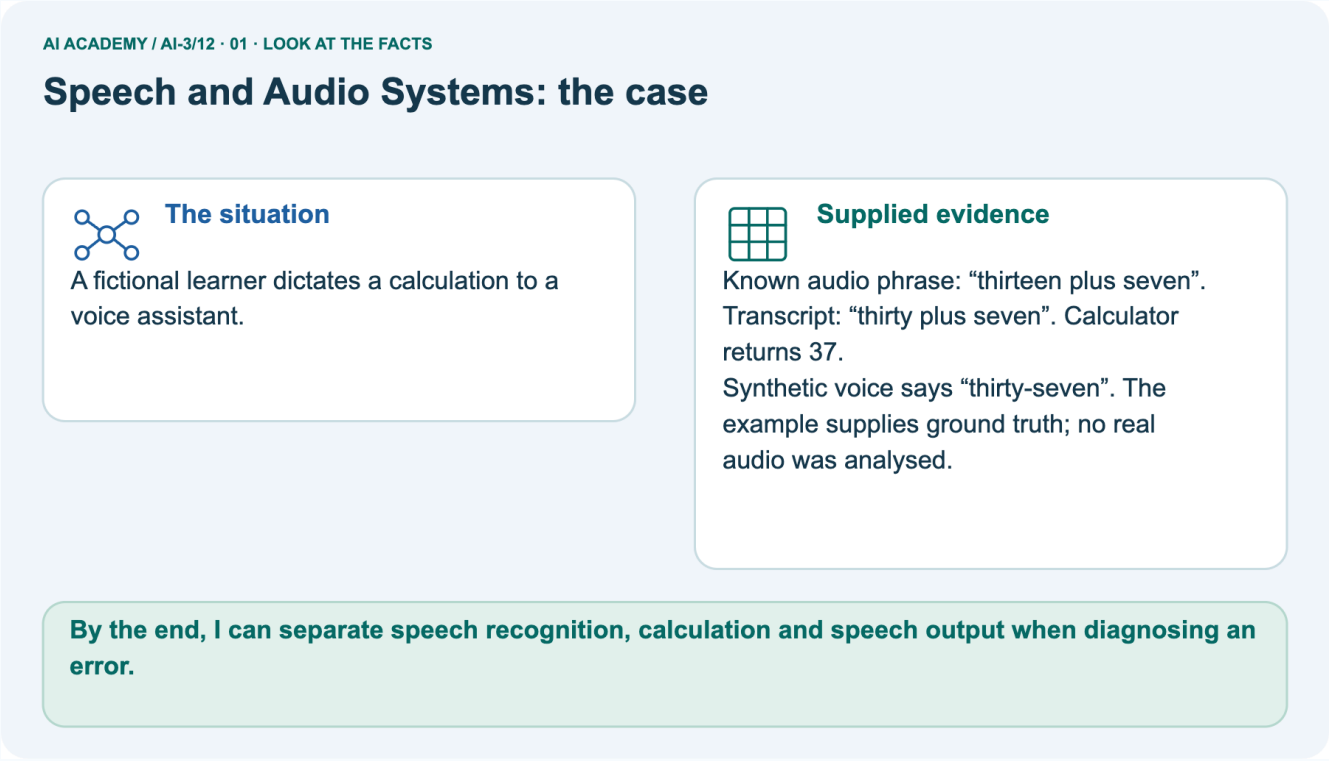
Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

# Visual concept journey

> Explain each Speech and Audio Systems picture from its labels, then record a relevant limit.

## 01. Speech and Audio Systems: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

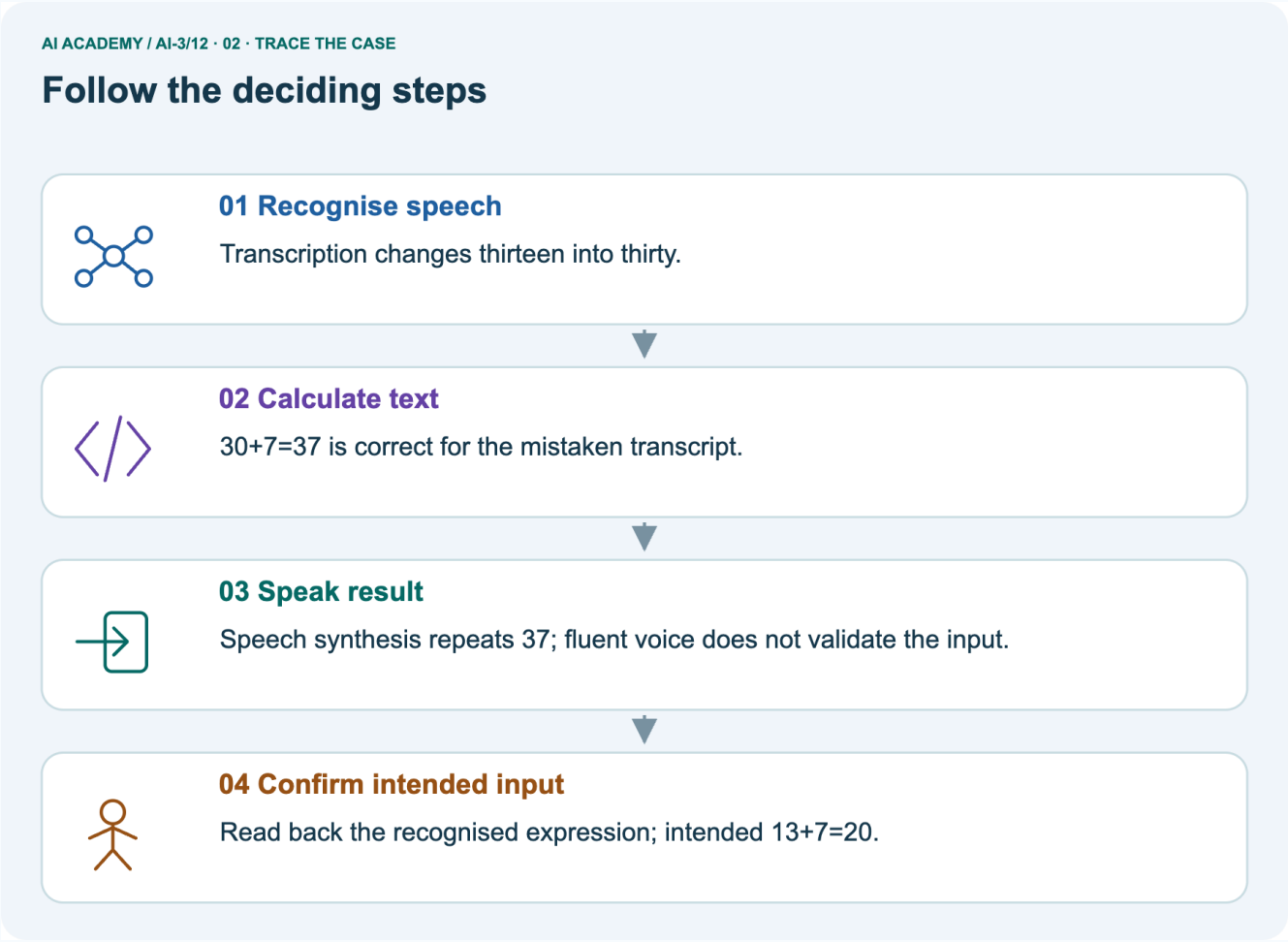
A fictional learner dictates a calculation to a voice assistant. Supplied facts: Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37. Synthetic voice says “thirty-seven”.

The example supplies ground truth; no real audio was analysed.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



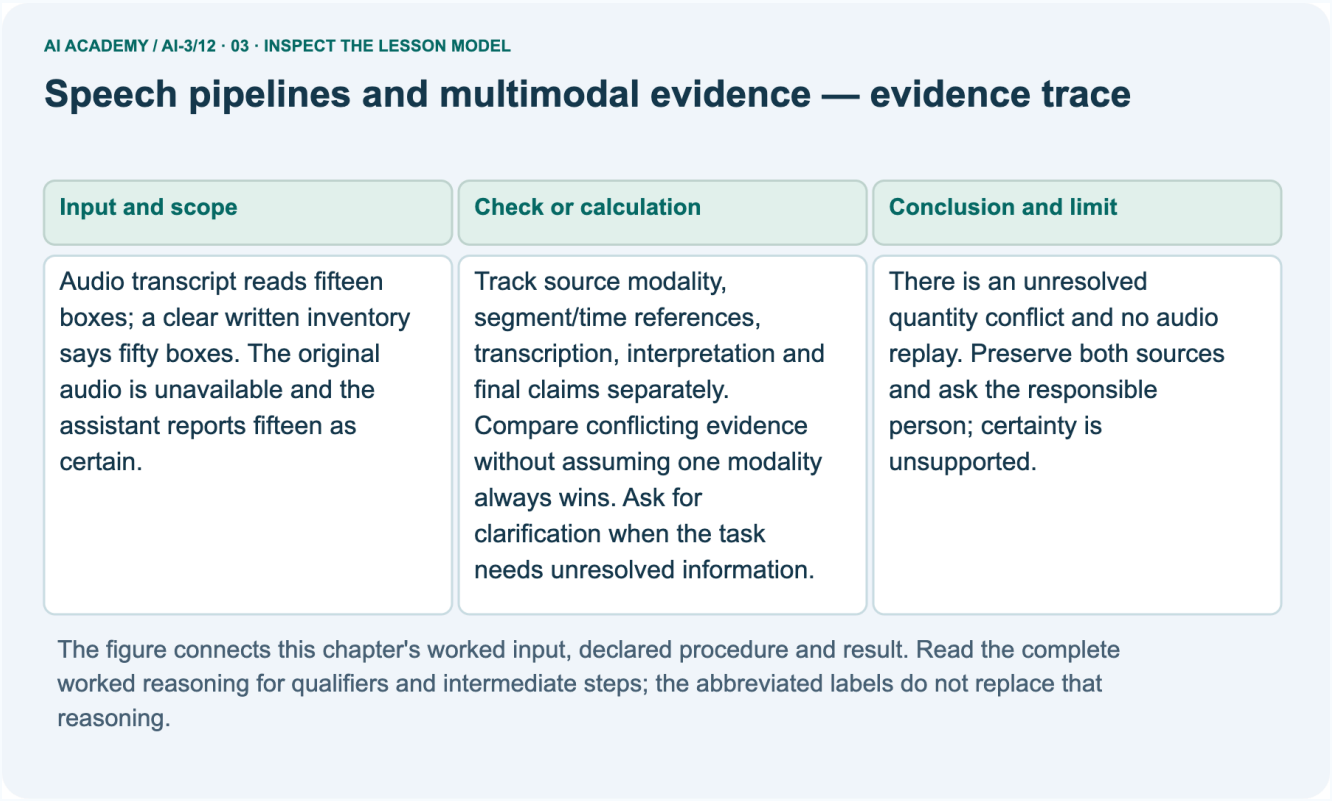

Follow the labelled picture with the matching explanation. Open the vector diagram

Recognise speech: **Transcription** changes thirteen into thirty. → Calculate text: 30+7=37 is correct for the mistaken transcript. → Speak result: Speech **synthesis** repeats 37; fluent voice does not validate the **input**. → Confirm intended **input**: Read back the recognised expression; intended 13+7=20.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

### 03. Speech pipelines and multimodal evidence — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

Speech pipelines and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

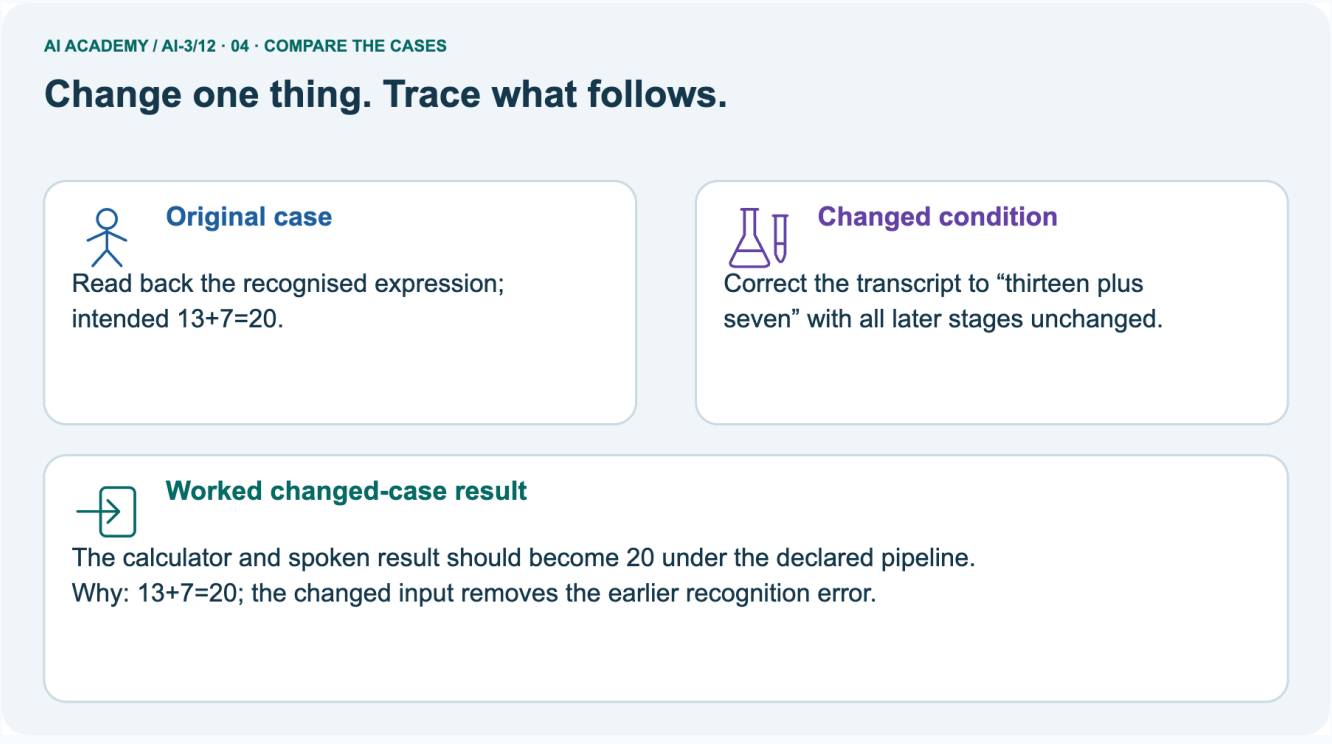
Compare conflicting **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity conflict and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

> Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

### 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

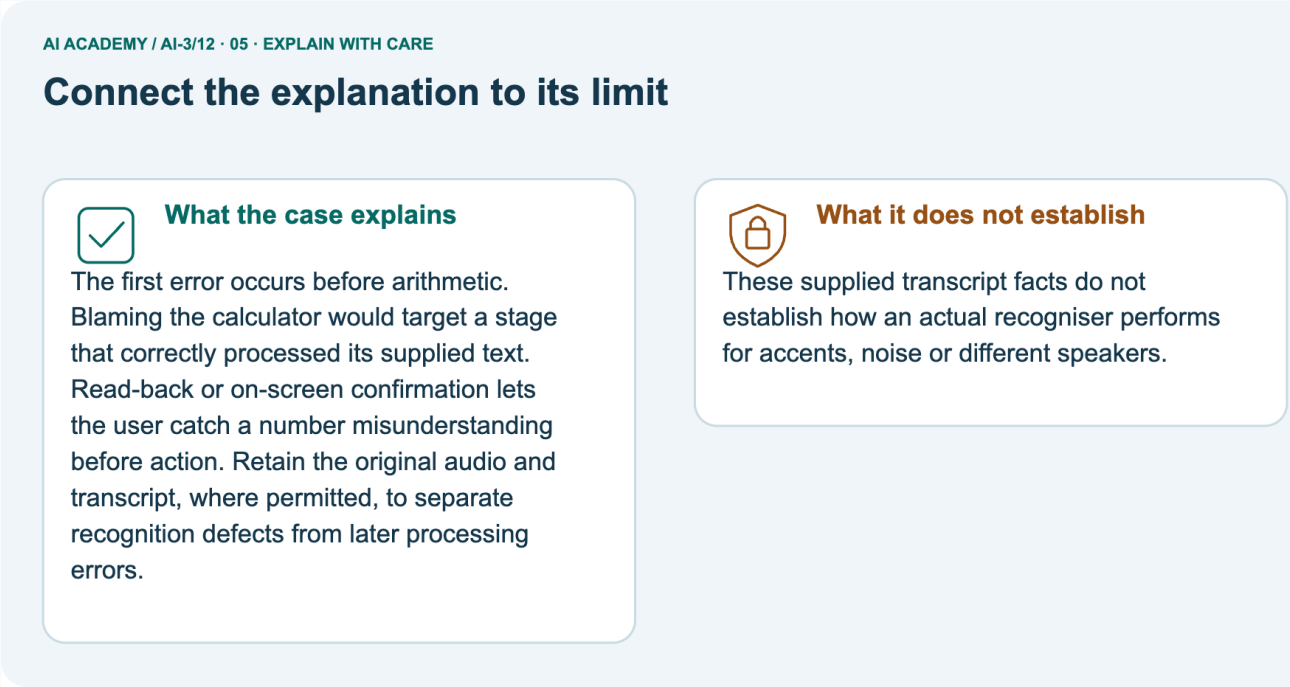
Original case: Read back the recognised expression; intended 13+7=20. Changed condition: Correct the transcript to “thirteen plus seven” with all later stages unchanged.

Expected result: The calculator and spoken result should become 20 under the declared pipeline. Reason: 13+7=20; the changed **input** removes the earlier recognition error.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

The first error occurs before arithmetic. Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

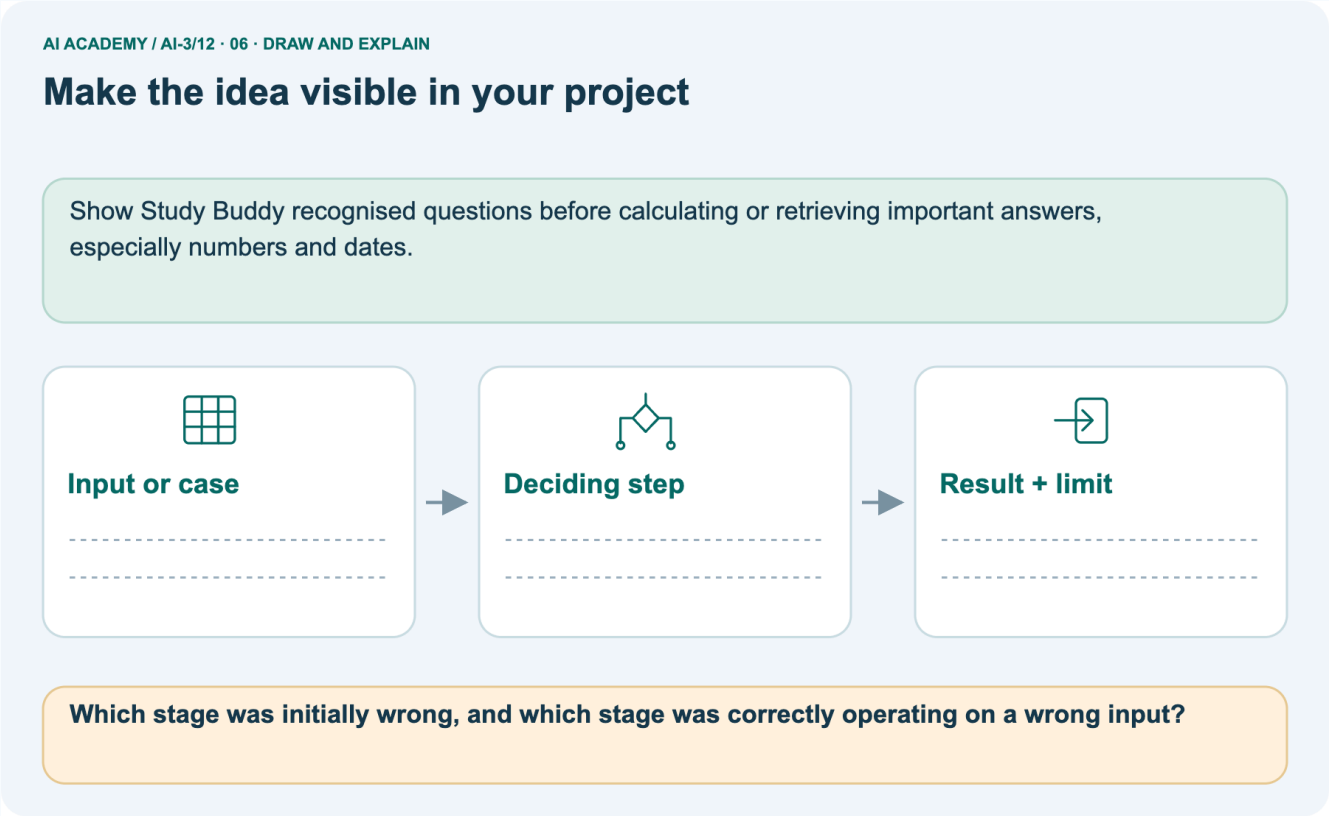
Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

Limit: These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

> Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates. Draw your **input**, deciding step, result and limit.

Supported question: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

> Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.